Radian measure

1

a $\frac{\pi}{180}$ **b** $\frac{180}{\pi}$

2

a 2π radians **b** π radians **c** $\frac{\pi}{2}$ radians **d** $\frac{5\pi}{4}$ radians

e $-\frac{5\pi}{3}$ radians **f** $\frac{28\pi}{45}$ radians **g** $\frac{29\pi}{180}$ radians **h** 2.81 radians

i 5.61 radians **j** $-\frac{\pi}{4}$ radians **k** $\frac{\pi}{4}$ radians **l** $\frac{11\pi}{6}$ radians

m $-\frac{\pi}{3}$ radians **n** $\frac{2\pi}{3}$ radians

3

a True

b False; 60° must equal to $\frac{\pi}{3}$ radians, because it is $\frac{1}{3}$ of 180°.

c False; 30° must equal to $\frac{\pi}{6}$ radians, because it is half of 60°.

d True

e True

f False; 225° must be equal to $\frac{5\pi}{4}$ radians, because it is five groups of 45°.

4

a 60° **b** 30° **c** 45° **d** 270°

e 225° **f** -120° **g** -300° **h** 240.6°

5

a

i 60° **ii** 150°

b

i $\frac{\pi}{4}$ radians **ii** $\frac{3\pi}{4}$ radians

c

i $30'$ **ii** $90^\circ 30'$

d

i $\frac{9\pi}{22}$ radians **ii** $\frac{10\pi}{11}$ radians

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Unit circle

6

a Quadrant 2 **b** Quadrant 3 **c** Quadrant 1 **d** Quadrant 4

7

a Point B **b** Point A **c** Point D **d** Point C

8

a Point A **b** Point B **c** Point C **d** Point D

9

a $\frac{1}{4}$ **b** $\frac{1}{3}$ **c** $\frac{2}{7}$ **d** $\frac{5}{2}$

10

a $\frac{7\pi}{6}, -\frac{5\pi}{6}$ **b** $\frac{\pi}{3}, -\frac{5\pi}{3}$ **c** $\frac{11\pi}{6}, -\frac{\pi}{6}$ **d** $\frac{2\pi}{3}, -\frac{4\pi}{3}$

11

a $\frac{14\pi}{13}$ **b** $-\frac{12\pi}{13}$

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